

Diagnostic Analysis of T-zone Enrichment Mechanisms

Abstract

This document summarizes a set of diagnostic simulation experiments designed to identify the mechanism responsible for spatial enrichment of observed cell deaths in a fixed T-shaped region, denoted by T . The main question is whether the observed T-zone enrichment is mainly caused by the candidate death intensity (λ_d), by ERK negative feedback, by the fixed-1000-death stopping rule, or by the localization of activation zones.

The results support a clear conclusion. The dominant mechanism controlling the existence of T-zone enrichment is activation localization. The candidate death intensity (λ_d) mainly changes the speed at which observed deaths accumulate, while ERK feedback modulates the strength of the enrichment. The fixed-1000-death stopping rule is not sufficient to explain the enrichment, because the enrichment remains visible under a fixed physical-time observation window.

1 Model and T-zone statistic

The spatial domain is

$$W = [0, L_x] \times [0, L_y], \quad L_x = L_y = 10.$$

The fixed T-shaped region T is composed of the middle vertical column of a 3×3 decomposition of the domain and the left middle arm. Its area fraction is

$$p_0 = \frac{|T|}{|W|} = 0.4444.$$

Therefore, under a spatially uniform null model, approximately 44.44% of observed deaths would be expected to fall inside the T-zone.

For each simulation run, the observed T-zone fraction is

$$\hat{\pi}_T = \frac{k_T}{n},$$

where k_T is the number of analyzed observed deaths inside the T-zone, and n is the number of analyzed observed deaths.

The T-zone density ratio is defined as

$$R_T = \frac{k_T/|T|}{(n - k_T)/(|W| - |T|)}.$$

A value $R_T > 1$ means that the density of observed deaths is higher inside the T-zone than outside it. A value $R_T = 1$ corresponds to equal death density inside and outside the T-zone.

The statistical test used for T-zone enrichment is the one-sided exact binomial test

$$H_0 : \pi_T = p_0, \quad H_A : \pi_T > p_0.$$

Under H_0 ,

$$K_T \sim \text{Binomial}(n, p_0),$$

and the one-sided p-value is

$$p_{\text{val}} = \mathbb{P}_{H_0}(K_T \geq k_T) = \sum_{j=k_T}^n \binom{n}{j} p_0^j (1-p_0)^{n-j}.$$

Since multiple parameter settings are tested, Bonferroni correction is applied. If m hypotheses are tested at familywise level $\alpha = 0.05$, then each individual test is compared to

$$\alpha_{\text{Bonf}} = \frac{0.05}{m}.$$

2 Spearman trend statistic

For the trend analysis, Spearman's rank correlation coefficient is used. Given paired observations

$$(x_i, y_i)_{i=1}^n,$$

the Spearman coefficient is the Pearson correlation coefficient computed after replacing the original values by their ranks:

$$\rho_s = \text{Corr}(\text{rank}(x_i), \text{rank}(y_i)).$$

Equivalently,

$$\rho_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2} \sqrt{\sum_{i=1}^n (S_i - \bar{S})^2}},$$

where

$$R_i = \text{rank}(x_i), \quad S_i = \text{rank}(y_i).$$

In this study, the Spearman statistic is used mainly as a descriptive trend indicator. Each trend test contains $n = 50$ observations, corresponding to five values of λ_d and ten independent simulation replicates per value. Because the associated p-value is an asymptotic approximation, it is not used as the central evidence. The main evidence is instead based on the T-zone fraction, the T-zone density ratio, the exact binomial enrichment test, and stability across independent replicates.

3 Experiment 1A: fixed 1000 analyzed deaths

Design

Experiment 1A uses the original stopping rule. After a burn-in of 500 observed deaths, the next 1000 observed deaths are analyzed. The activation regime is fixed at

$$\lambda_{a,T} = 0.5, \quad \lambda_{a,c} = 0.005, \quad \beta_{a,R} = 2.5, \quad \beta_{a,T} = 3.0.$$

The candidate death intensity is swept over

$$\lambda_d \in \{0.25, 0.5, 1.0, 2.0, 4.0\}.$$

Three ERK regimes are tested:

$$\text{strong ERK protection: } (\beta_{d,R}, \beta_{d,T}) = (1.0, 0.4),$$

$$\text{reference ERK: } (\beta_{d,R}, \beta_{d,T}) = (2.0, 0.8),$$

$$\text{weak ERK protection: } (\beta_{d,R}, \beta_{d,T}) = (4.0, 1.6).$$

For each parameter setting, ten independent replicates are simulated.

Results for strong ERK protection

For strong ERK protection, the mean analysis time decreases from

$$291.365 \pm 14.244$$

at $\lambda_d = 0.25$ to

$$117.284 \pm 6.705$$

at $\lambda_d = 4.0$. This shows that increasing λ_d strongly accelerates the accumulation of the 1000 analyzed deaths.

The T-zone fraction remains above the null area fraction ($p_0 = 0.4444$) for all values of λ_d . The replicate-level T-zone fractions are

$$0.6083 \pm 0.0205 \quad (\lambda_d = 0.25),$$

$$0.5990 \pm 0.0269 \quad (\lambda_d = 0.5),$$

$$0.5944 \pm 0.0226 \quad (\lambda_d = 1.0),$$

$$0.5838 \pm 0.0242 \quad (\lambda_d = 2.0),$$

and

$$0.5589 \pm 0.0232 \quad (\lambda_d = 4.0).$$

The corresponding mean density ratios range from approximately 1.591 to 1.949. All five pooled binomial tests are significant after Bonferroni correction.

The Spearman trend between λ_d and analysis time is strongly negative:

$$\rho_s(\lambda_d, \text{analysis time}) = -0.966.$$

This confirms the monotone decrease of analysis time as λ_d increases. The trend between λ_d and T-zone fraction is also negative in this regime,

$$\rho_s(\lambda_d, \hat{\pi}_T) = -0.532,$$

but the enrichment remains present at every tested value of λ_d .

Results for the reference ERK regime

For the reference ERK regime, the mean analysis time decreases from

$$172.814 \pm 8.532$$

at $\lambda_d = 0.25$ to

$$33.617 \pm 3.503$$

at $\lambda_d = 4.0$. Again, λ_d mainly controls the speed of death accumulation.

The T-zone fractions are

$$0.6424 \pm 0.0139 \quad (\lambda_d = 0.25),$$

$$0.6395 \pm 0.0255 \quad (\lambda_d = 0.5),$$

$$0.6295 \pm 0.0317 \quad (\lambda_d = 1.0),$$

$$0.6219 \pm 0.0357 \quad (\lambda_d = 2.0),$$

and

$$0.6268 \pm 0.0512 \quad (\lambda_d = 4.0).$$

These values are all clearly above ($p_0 = 0.4444$). The corresponding mean density ratios range from approximately 2.087 to 2.250. All five pooled binomial tests are significant after Bonferroni correction.

The Spearman trend between λ_d and analysis time is

$$\rho_s(\lambda_d, \text{analysis time}) = -0.974,$$

whereas the trend between λ_d and T-zone fraction is weak:

$$\rho_s(\lambda_d, \hat{\pi}_T) = -0.200.$$

Thus, in the reference ERK regime, increasing λ_d strongly reduces the time required to obtain 1000 deaths, but it does not strongly change the T-zone fraction.

Results for weak ERK protection

For weak ERK protection, the mean analysis time decreases from

$$150.154 \pm 8.422$$

at $\lambda_d = 0.25$ to

$$12.441 \pm 1.605$$

at $\lambda_d = 4.0$. This is the fastest regime among the three ERK settings at high λ_d .

The T-zone fractions are

$$0.6584 \pm 0.0202 \quad (\lambda_d = 0.25),$$

$$0.6525 \pm 0.0199 \quad (\lambda_d = 0.5),$$

$$0.6499 \pm 0.0343 \quad (\lambda_d = 1.0),$$

$$0.6171 \pm 0.0576 \quad (\lambda_d = 2.0),$$

and

$$0.6370 \pm 0.0411 \quad (\lambda_d = 4.0).$$

All values remain above the geometric null ($p_0 = 0.4444$). The mean density ratios range from approximately 2.105 to 2.421. All five pooled binomial tests are significant after Bonferroni correction.

The Spearman trend between λ_d and analysis time is

$$\rho_s(\lambda_d, \text{analysis time}) = -0.980.$$

The trend between λ_d and T-zone fraction is

$$\rho_s(\lambda_d, \hat{\pi}_T) = -0.371.$$

Conclusion from Experiment 1A

Experiment 1A shows that increasing λ_d strongly reduces the physical time required to obtain 1000 analyzed deaths. This is visible in all ERK regimes through strongly negative Spearman correlations between λ_d and the analysis time.

However, the T-zone fraction remains systematically above the null area fraction ($p_0 = 0.4444$). Across all 15 parameter settings, the pooled T-zone fraction ranges from

$$0.5589 \quad \text{to} \quad 0.6584,$$

and the pooled density ratio ranges from

$$1.584 \quad \text{to} \quad 2.409.$$

All 15 pooled binomial tests are significant after Bonferroni correction.

Therefore, under the fixed-1000-death stopping rule, T-zone enrichment is robust across the tested death and ERK parameters. The parameter λ_d mainly controls the speed of the observed death process rather than being the primary source of T-zone enrichment.

4 Experiment 1B: fixed physical-time control

Design

Experiment 1B is designed to check whether T-zone enrichment is an artefact of the fixed-1000-death stopping rule. Instead of stopping after 1000 analyzed deaths, the simulation observes deaths over a fixed physical time window after burn-in. This observation window is estimated from Experiment 1A at the reference parameter setting

$$\lambda_d = 1.0, \quad \beta_{d,R} = 2.0, \quad \beta_{d,T} = 0.8.$$

The resulting physical observation window is

$$T_{\text{obs}} = 61.349906.$$

All λ_d values and ERK regimes are then compared over this same physical time window.

Results for strong ERK protection

Under strong ERK protection, the mean number of analyzed deaths increases with λ_d :

$$219.5 \pm 36.5 \quad (\lambda_d = 0.25),$$

$$285.5 \pm 40.6 \quad (\lambda_d = 0.5),$$

$$346.3 \pm 35.2 \quad (\lambda_d = 1.0),$$

$$455.0 \pm 37.9 \quad (\lambda_d = 2.0),$$

and

$$551.8 \pm 42.4 \quad (\lambda_d = 4.0).$$

Thus, over the same physical time, larger λ_d produces more observed deaths.

The T-zone fractions remain above ($p_0 = 0.4444$):

$$0.6147 \pm 0.0624, \quad 0.5812 \pm 0.0419, \quad 0.5840 \pm 0.0324, \quad 0.5792 \pm 0.0294, \quad 0.5912 \pm 0.0294.$$

The pooled density ratios range from approximately 1.718 to 1.937. All five pooled binomial tests are significant after Bonferroni correction.

Results for the reference ERK regime

Under the reference ERK regime, the mean number of observed deaths increases from

$$360.7 \pm 27.9$$

at $\lambda_d = 0.25$ to

$$1918.9 \pm 147.6$$

at $\lambda_d = 4.0$.

The T-zone fractions are

$$0.6492 \pm 0.0337, \quad 0.6427 \pm 0.0297, \quad 0.6317 \pm 0.0330, \quad 0.6197 \pm 0.0241, \quad 0.6076 \pm 0.0256.$$

These values are all above ($p_0 = 0.4444$). The pooled density ratios range from approximately 1.929 to 2.309. All five pooled binomial tests are significant after Bonferroni correction.

The Spearman trend between λ_d and the number of observed deaths is strongly positive:

$$\rho_s(\lambda_d, n_{\text{deaths}}) = 0.977.$$

Results for weak ERK protection

Under weak ERK protection, the mean number of observed deaths increases very strongly:

$$420.9 \pm 44.3 \quad (\lambda_d = 0.25),$$

$$836.2 \pm 91.2 \quad (\lambda_d = 0.5),$$

$$1540.6 \pm 86.5 \quad (\lambda_d = 1.0),$$

$$2636.3 \pm 269.8 \quad (\lambda_d = 2.0),$$

and

$$4581.9 \pm 180.0 \quad (\lambda_d = 4.0).$$

The T-zone fractions remain high:

$$0.6599 \pm 0.0486, \quad 0.6500 \pm 0.0275, \quad 0.6407 \pm 0.0218, \quad 0.6491 \pm 0.0226, \quad 0.6396 \pm 0.0222.$$

The pooled density ratios range from approximately 2.215 to 2.406. All five pooled binomial tests are significant after Bonferroni correction.

Conclusion from Experiment 1B

Experiment 1B confirms that the T-zone enrichment is not an artefact of the fixed-1000-death stopping rule. When all parameter settings are observed over the same physical time window, increasing λ_d mainly increases the number of observed deaths. Nevertheless, the T-zone fraction remains consistently above the null area fraction ($p_0 = 0.4444$).

Across all 15 fixed-time settings, the pooled T-zone fraction ranges from

$$0.5789 \quad \text{to} \quad 0.6581,$$

and the pooled density ratio ranges from

$$1.718 \quad \text{to} \quad 2.406.$$

All 15 pooled binomial tests remain significant after Bonferroni correction. Therefore, T-zone enrichment is present in real physical time and is not created merely by waiting until 1000 deaths have occurred.

5 Experiment 2: activation localization and death/ERK robustness

Design

Experiment 2 tests whether activation localization controls the existence of T-zone enrichment. The T-zone activation contrast is fixed:

$$\lambda_{a,T} = 0.5, \quad \lambda_{a,c} = 0.005.$$

Three activation regimes are compared:

$$\text{non-local activation: } (\beta_{a,R}, \beta_{a,T}) = (1.0, 1.2),$$

$$\text{medium activation: } (\beta_{a,R}, \beta_{a,T}) = (2.5, 3.0),$$

$$\text{local activation: } (\beta_{a,R}, \beta_{a,T}) = (5.0, 3.0).$$

Inside each activation regime, the same λ_d values and ERK regimes are swept. This produces 15 death/ERK parameter settings per activation regime.

Non-local activation

In the non-local activation regime, no setting is significant after Bonferroni correction. The pooled T-zone fraction ranges from

$$0.3865 \quad \text{to} \quad 0.4474,$$

with a mean pooled fraction of

$$0.4236.$$

The pooled density ratio ranges from

$$0.787 \quad \text{to} \quad 1.012,$$

with a mean density ratio of

$$0.921.$$

These values are close to, or even below, the null area fraction ($p_0 = 0.4444$). Therefore, non-local activation does not produce robust T-zone enrichment.

At the replicate level, the mean T-zone fractions across parameter settings range from

$$0.3865 \quad \text{to} \quad 0.4474,$$

with standard deviations ranging from approximately

$$0.0142 \quad \text{to} \quad 0.0411.$$

This confirms that the absence of enrichment is stable across independent replicates.

Medium activation

In the medium activation regime, all 15 settings are significant after Bonferroni correction. The pooled T-zone fraction ranges from

$$0.5686 \quad \text{to} \quad 0.6574,$$

with a mean pooled fraction of

$$0.6220.$$

The pooled density ratio ranges from

$$1.648 \quad \text{to} \quad 2.399,$$

with a mean density ratio of

$$2.071.$$

Thus, medium activation produces clear and robust T-zone enrichment.

At the replicate level, the mean T-zone fractions across parameter settings range from

$$0.5686 \quad \text{to} \quad 0.6574,$$

with standard deviations ranging from approximately

$$0.0128 \quad \text{to} \quad 0.0648.$$

This shows that the enrichment is not only significant in pooled counts but also stable across independent simulation replicates.

Local activation

In the local activation regime, all 15 settings are significant after Bonferroni correction. The pooled T-zone fraction is extremely high, ranging from

$$0.8950 \quad \text{to} \quad 0.9136,$$

with a mean pooled fraction of

$$0.9068.$$

The pooled density ratio ranges from

$$10.655 \quad \text{to} \quad 13.218,$$

with a mean density ratio of

$$12.211.$$

Thus, local activation produces very strong T-zone enrichment.

At the replicate level, the mean T-zone fractions across parameter settings range from

$$0.8950 \quad \text{to} \quad 0.9136,$$

with standard deviations ranging from approximately

$$0.0050 \quad \text{to} \quad 0.0166.$$

This is the most stable and strongest enrichment regime.

Conclusion from Experiment 2

Experiment 2 is the most decisive diagnostic experiment. When activation zones are non-local, T-zone enrichment disappears. When activation zones are medium or local, T-zone enrichment appears systematically across all death/ERK settings.

This demonstrates that activation localization is the dominant mechanism controlling the existence of T-zone enrichment. Death and ERK parameters can modulate the strength and speed of the process, but they do not create robust enrichment in the absence of localized activation.

Overall interpretation The three experiments separate three possible explanations.

First, Experiment 1A shows that T-zone enrichment is robust under the original fixed-1000-death stopping rule. All ERK regimes and all tested values of λ_d produce pooled T-zone fractions above the null area fraction and significant one-sided binomial tests after Bonferroni correction.

Second, Experiment 1B shows that the enrichment is not caused by the stopping rule. When the physical observation time is fixed instead of the number of deaths, T-zone enrichment remains present. Therefore, the enrichment reflects a genuine spatial bias in the simulated death process rather than an artefact of stopping after exactly 1000 deaths.

Third, Experiment 2 shows that activation localization controls whether T-zone enrichment exists. Non-local activation gives no significant enrichment, medium activation gives consistent enrichment, and local activation gives extremely strong enrichment.

The final mechanistic interpretation is therefore:

activation localization is the main driver of T-zone enrichment.

The parameter λ_d mainly controls the temporal speed of the death process:

$$\lambda_d \uparrow \implies \text{analysis time decreases under fixed death count,}$$

and

$$\lambda_d \uparrow \implies \text{number of observed deaths increases under fixed physical time.}$$

ERK feedback modulates the strength of enrichment, but it does not determine whether enrichment exists.

6 result

The diagnostic experiments indicate that T-zone enrichment is not an artefact of the fixed-1000-death stopping rule. Under the fixed-death design, all tested death/ERK settings show significant enrichment relative to the geometric null ($p_0 = |T|/|W| = 0.4444$). Under a fixed physical-time observation window, the enrichment remains present, while λ_d mainly controls the number of observed deaths. The strongest evidence comes from the activation-localization experiment: non-local activation produces no significant enrichment, whereas medium and local activation produce significant enrichment in all tested death/ERK settings. Therefore, activation localization is the dominant mechanism controlling the existence of T-zone enrichment, while λ_d primarily controls speed and ERK feedback modulates enrichment strength.